

Marco Castellari

marco.castellari01@gmail.com (+39) 342-8838063
16/09/1993 – Imola (BO)

PROFILE

Versatile Electronics and Telecommunications Engineer with expertise in firmware and software design and development for high-performance systems in motorsport, industrial automation and medical devices. Adaptable team player with proven ability to deliver under pressure, with a passion for new challenges and continuous learning.

WORK EXPERIENCE

R&D Senior Software Engineer Oct 2024 – Present
Cefla S.C., Imola (IT)

- Software architecture and real-time C++ framework development for X-ray imaging systems
- Design and implementation of medical imaging reconstruction algorithms

R&D HW/FW Engineer May 2023 – Sept 2024
Mectronik s.r.l., Imola (IT)

- Firmware development for high-performance motorsport applications and collaboration in hardware design
- Trackside engineering support for race strategy and engine map optimization

R&D Software Engineer Apr 2018 – May 2023
SACMI Innovation Lab, Imola (IT)

- Development of C++/GPU and FPGA solutions for high-performance and real-time computing
- Maintenance of legacy C applications and on-site prototype commissioning

EDUCATION

State examination with qualification for the profession of Information Engineer 2021
Alma Mater Studiorum - Università di Bologna

MSc Electronics & Telecommunications Eng. 2015–2018
Università degli Studi di Ferrara 110/110 *cum Laude*
Thesis: *Performance Evaluation of Emerging FPGA Technologies*

BSc Electronics & Computer Science Eng. 2012–2015
Università degli Studi di Ferrara 110/110
Thesis: *Localization Systems based on LTE Signals of Opportunity*

High School Diploma 2007 - 2012
ITIS - G. Marconi, Lugo (RA) 100/100 *cum Laude*
QUALIFIED INDUSTRIAL TECHNICIAN
Specialization in Electronics and Telecommunications

RESEARCH EXPERIENCE

Fermilab Summer Internship July – September 2017
Fermi National Accelerator Laboratory, Batavia (USA, IL)
Novel RF Tuning of Jacketed Multi-Cell SRF Cavities

MPSoC Research Group August – September 2015
Università degli Studi di Ferrara
Optical Interconnection Networks for HPC

SKILLS

Programming C/C++, C#, CUDA, Python, MATLAB, VHDL

Hardware STM32, ESP32, NXP, Arduino, FPGA, CAN bus, Lauterbach, Segger, Laboratory Instrumentation

Tools Git/SVN, Docker, Jenkins, Altium Designer, Vivado, Quartus

Architecture High-Performance Computing, Real-Time and Embedded Systems, Residuality Theory in software architecture

Soft Skills Passion for learning, Care for details, Teamwork

Languages Italian (Native), English (B2)

INTERESTS

Electronics for motorsport (suspension testbench and telemetry system) • Mechanics and engines • Firmware/Software conference attendance • Travelling

PUBLICATION

Tuning of Multicell Superconducting Accelerating Cavities using Pressurized Balloons, 2022 IEEE Transactions on Applied Superconductivity.

Correlating Power Efficiency and Lifetime to Programming Strategies in RRAM-Based FPGAs, 2018 New Generation of CAS (NGCAS).

Populating and exploring the design space of Wavelength-Routed Optical Network-on-Chip topologies by leveraging the add-drop filtering primitive, 2016 Tenth IEEE/ACM International Symposium on Networks-on-Chip (NOCS).